INDOSAW
COMMITTED TO EXCELLENCE

THERMAL AND ELECTRICAL CONDUCTIVITY OF METAL

Instruction Manual



Note- The temps. across the copper or aluminium rod at different points can be measured either by using Digital thermometers (SK068) or temp. sensor using data-logger (SK197).

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INTRODUCTION:

Thermal conductivity of materials can't be measured directly unlike electrical conductivity. There are two different types of techniques for measurement of thermal conductivity, namely steady-state techniques and transient techniques. In general, steady-state techniques perform a measurement when the temperature of the materials measured does not change with time. This makes signal analysis simple. The transient techniques perform a measurement during the process of heating up and it is not simple. The advantage is that measurements can be made quickly.

In metals, thermal conductivity approximately tracks electrical conductivity according to Wiedmann-Franz law, as freely moving electrons transfer not only electric current but also heat energy. However, the general correlation between electrical and thermal conductivity does not hold for other materials, due to the increased importance of phonon carriers for heat in nonmetals.

OBJECTIVE:

1. Determine the heat capacity of the calorimeter in a mixture experiment as a preliminary test. Measure the calefaction of water at a temperature of 0°C in a calorimeter due to the action of the ambient temperature as a function of time for copper and aluminum.
2. To begin with, establish a constant temperature gradient in a metal rod with the use of two heat reservoirs (boiling water and ice water).
3. After removing the pieces of ice, measure the calefaction of the cold water as a function of time and determine the thermal conductivity of the metal rod (Copper/Aluminium**)**.
4. To determine the electrical conductivity of aluminium and Copper by recording a current-voltage characteristic curve.
5. Test of the Wiedemann-Franz law and find out the Lorentz number.

LIST OF ITEMS:

S. No.	ITEM NAME	QTY.
1	Conductivity Rod Al	1
2	Conductivity Rod Cu	1
3	Upper Calorimeter Vessel with Lid	1
4	Lower Calorimeter Vessel with Stirrer	1
5	Laboratory Jack	1
6	Weighing Balance with adaptor	1
7	Immersion Heater	1
8	Digital Thermometer	3
9	Heat Conductive Paste	2
10	Universal Clamp	2
11	Base	1
12	Support Rod	1
13	Bosshead	2
14	Power Supply 0-30V, 20amp	1
15	Microvoltmeter	1
16	Connecting lead 100cm Red and Black Pair	1
17	Connecting lead 50cm Red and Black Pair	1
18	Glass Beaker 500ml	1
19	Gloves (pair)	1
20	Syringe Plastic	1
21	Plastic Beaker 2000ml	1
22	Stop Watch	1
23	Power Cord	1

THEORY AND EVALUATION

If a temperature difference exists between different locations of a body, heat conduction occurs. In this experiment there is a one-dimensional temperature gradient along a rod. The quantity of heat dQ transported with time dt is a function of the cross-sectional area A and the temperature gradient ΔT/Δx perpendicular to the surface.

$$\frac{dQ}{dt} = -\lambda A.\frac{\delta T}{\delta x} \qquad \text{----------1}$$

is the heat conductivity of the substance.

The temperature distribution in a body is generally a function of location and time and is in accordance with the Boltzmann transport equation

$$\frac{\Delta Q}{\Delta t} = \frac{\lambda}{\rho.c}.\frac{\delta^2 T}{\delta x^2} \text{----------2}$$

Where ρ is the density and 'c' is the specific heat capacity of the substance.

After a time, a steady state

$$\frac{\delta T}{\delta x} = 0 \text{ ----------3}$$

is achieved if the two ends of the metal rod having a length l are maintained at constant temperatures T_1 and T_2, respectively, by two heat reservoirs.

On reaching constant temperature gradient (steady state), equation 3 can be considered as having been satisfied by the metal rod. In order to calculate the heat energy transported by the metal rod according to Equation 1, the ambient heat fraction must be subtracted.

That is

$$\frac{dQ_{rod}}{dt} = \frac{dQ_{total}}{dt} - \frac{dQ_{surrounding}}{dt} \qquad \text{----------4}$$

At room temperature the conduction electrons in metal have a much greater mean free path than the phonons. For this reason heat conduction in metal is primarily due to the electrons.

The correlation between the thermal conductivity λ and the electrical conductivity σ is established by the Wiedemann-Franz law:

$$\frac{\lambda}{\sigma} = LT \qquad \text{----------5}$$

The Lorenz number L, which can be experimentally determined using Equation (5), is established by the theory of electron vapour (for temperatures above the Debye temperature) to be:

$$L = \frac{\pi^2}{3}.\frac{k^2}{e^2} = 2.4x10^{-8}\ \frac{W\Omega}{K^2} \qquad \text{----------6}$$

Where k = Universal gas constant = $1.38x10^{-23}$ J/K,

e = Elementary unit charge = $1.602x10^{-19}$ AS

PROCEDURE FOR THERMAL CONDUCTIVITY ESTIMATION:

Heat capacity:- It is the amount of heat required to change its temperature by 1°

Q = cdt

Q = heat supplied (J)

C = heat capacity (Jk^{-1})

1) **Measurement of heat capacity of lower calorimeter**

The heat capacity of the calorimeter is obtained from results of the mixing experiment and the following formula:

$$c = c_w.m_w.\frac{T_w - T_m}{T_m - T_R} \qquad \text{----------7}$$

c_w = Specific heat capacity of water, m_W = Mass of the water, T_w = Temperature of the hot water, T_m = Mixing temperature and T_R = Room temperature

Method:

1. Weigh the calorimeter at room temperature.
2. Measure the room temperature and temperature of preheated water provided.
3. Fill the calorimeter with hot water, determine the mixing temperature of water.
4. Reweigh the calorimeter to determine the mass of the water.
5. Calculate the heat capacity of the calorimeter using eq^{n} -7.

2) Determination of the influence of the surroundings

The addition of heat from the surroundings is calculated from the temperature increase (Temperature (T) rise of the cold water in the calorimeter)

$$Q = (c_w . m_w + c).(T - T_0) \text{ ----------8}$$

Where T_0 = Temperature at time t = 0 sec.
*By plotting a graph between heat energy supplied versus time, heat energy supplied by surroundings with time **(dQsurr/dt)** can be estimated.*

Method:

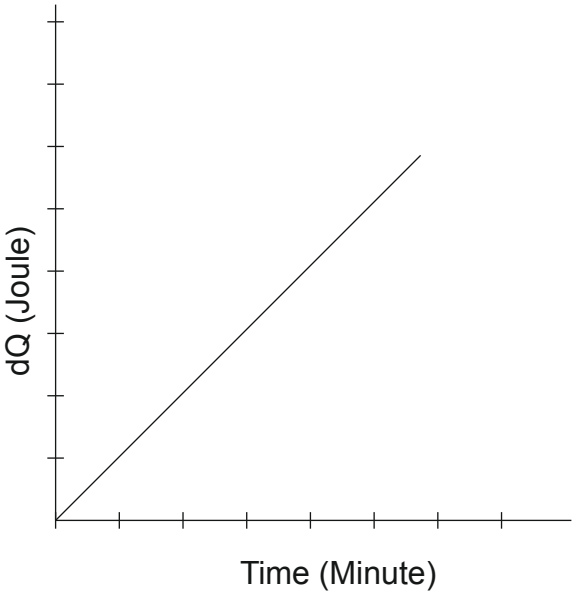
1. Weigh the empty lower calorimeter.
2. Add ice to the lower calorimeter till the temperature reaches to 0°C.
3. Then remove the ice and take the temperature readings in 1 min interval till the temperature rises to 12°C.
4. Reweigh the calorimeter to determine the mass of the water that it contains.



Table for calefaction of cold water using eq^{n} - 8

Time (m)	Temp. °C	dQsurr (J)

Slope of the graph give us $\frac{dQ}{dt}$ Surr



3) Determination of the heat flow through metal rod thermal conductivity

Equation for the measurement of the thermal conductivity of Aluminium/copper rod is given by

$$\frac{dQ_{rod}}{dt} = -\lambda A . \frac{\Delta T}{\Delta x}$$

Method:

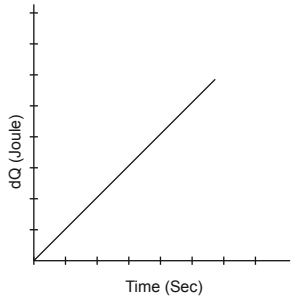
1. Perform the experimental set-up according to Fig. 1.
2. Weigh the empty, lower calorimeter.
3. Insert the insulated end of the metal rod into the upper calorimeter vessel. To improve the heat transfer, cover the end of the metal rod with heat-conduction paste.
4. Attach the metal rod to the support stand in such a manner that the lower calorimeter can be withdrawn from beneath it.
5. The height of the lower calorimeter can be changed with laboratory jack. When doing so, care must be taken to ensure that the non-insulated end of the rod remains completely immersed in the cold water during the experiment.
6. The temperature probe must be positioned as close to the rod as possible.
7. The outermost indentations on the rod are used to measure the temperature difference in the rod. To improve the heat transfer between the rod and the temperature probe, use heat conduction paste.
8. Using an immersion heater, bring the water in the upper calorimeter to a boil, and keep it at this temperature.
9. Ensure that the upper calorimeter is well filled to avoid a drop in temperature due to contingent refilling with water.
10. Keep the water in the lower calorimeter at 0°C with the help of ice.
11. The measurement can be begun when a constant temperature gradient has been established between the upper and lower surface probes.
12. At the onset of measurement, remove the ice from the lower calorimeter.
13. Measure and record the change in the differential temperature and the temperature of the water in the lower calorimeter for a period of 5 minutes.
14. Weigh the water-filled calorimeter and determine the mass of the water.
15. Sample Calculations.

Time	Temp. of lower calorimeter °C	ΔT °C	ΔQ (J)

where

$$\Delta Q = (c_w m_w + c)\ (T - T_0))$$

Slope of graph give $\frac{dQ}{dt}$ total



So

$$\left(\frac{dQ}{dt}\right)_{rod} = \left(\frac{dQ}{dt}\right)_{total} - \left(\frac{dQ}{dt}\right)_{surr}$$

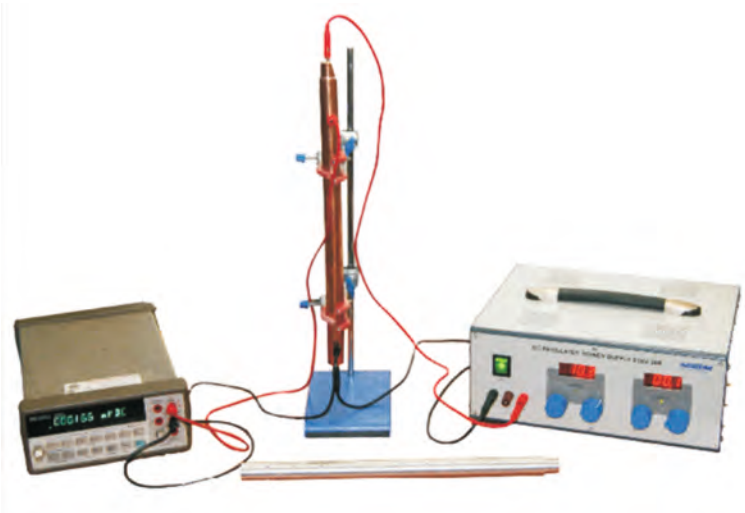
Calculate λ using eqⁿ 6

Electrical conductivity theory:

Ohm's Law $V \alpha I$ **So** $R = \frac{V}{I}$

and $R = \frac{\rho L}{A} = \frac{1}{\sigma}\frac{L}{A}$ } **ρ =Resistivity of material**

$\sigma = \frac{LA}{R}$ } **σ =Conductivity**



B. Measurement of the electrical conductivity.

a. Perform the experimental set-up according to the circuit diagram in Fig. 2 (set-up in accordance with a 4-conductor measuring method). b. Set the Microvoltmeter 2000μVolt range. Offset the display reading.

c. Slowly increase the current value through high current power supply.

d. Read and note the values for current and voltage, L and ρ

e. σ = L/AR

L of Cu and Al rod = 42.5 cm
Dia, D = 25 mm

Material	**V (Microvolt)**	**I (Amp)**	$\sigma = \frac{L}{AR} = \frac{LI}{AV}$ $(ohm^{-1}\ m^{-1})$
Al	76	3	3.42×10^7
Cu	48	3	5.37×10^7

From the graph slope will give us R

So electrical conductivity $\sigma = \frac{L}{AR}$

L = length of rod between to probes.
A = πr^2 area of crossection of hot end.

Relation between electrical and thermal conductivity

Wiedemann-FRANZ law

At a given temperature the thermal and electrical conductivity of metals are proportional, but raising the temperature increase the thermal conductivity while decreasing the electrical conductivity.

This behavior is quantified in the Wiedemann-FRANZ law.

$$\frac{\lambda}{\sigma} = LT \qquad \text{or} \qquad L = \frac{\lambda}{\sigma T}$$

Theoretical value is given in eq 6

Results

Heat capacity of calorimeter = ...J/k

Thermal conductivity of Al= 3.42×10^{7} $wm^{-1}k^{-1}$

Literature value = 205 $wm^{-1}k^{-1}$

Electrical conductivity of Al= ..... $\Omega^{-1}m^{-1}$

Literature value = 3.75×10^{7} $\Omega^{-1}m^{-1}$

Thermal conductivity of Cu =5.33×10^{-7} $wm^{-1}k^{-1}$

Literature value = 386 $wm^{-1}k^{-1}$

Electrical conductivity of Cu= ..... $\Omega^{-1}m^{-1}$

Literature value = 5.882×10^{7} $\Omega^{-1}m^{-1}$

=5.3×10^{7} $\Omega^{-1}m^{-1}$

PRECAUTIONS:

A. The mixing temperature is close to the temperature of hot water, so one has to be prompt in reading the fall in temperature versus time.
B. Care should be taken to ensure the immersion heater used is not exposed to air for long.
C. Always maintain minimum water level in the upper calorimeter while maintaining at 100 °C.
D. Handle temperature probes with care. Bending front portion too much may damage the sensor.

QUESTIONS:

A. What is the difference between specific heat and heat capacity? Which one of them is an extensive variable?
B. What is the necessity of maintaining steady state for thermal conductivity measurement?
C. What is the necessity of thermal conducting paste? What is thermal contact resistance?

Calculated values for electrical conductivity are very close to the literature values but for the thermal conductivity it is significantly different, because.

1. It is very difficult to obtain constant temperature gradient since we have to keep both the calorimeter at fixed temperatures.
2. Even though after coming to steady state the system will get disturbed, as no insulator and conductor is perfect.